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GitHub → https://github.com/Sagarwal-bit/cloud-command-parser

Learning and Tasks

Intro

In this project I developed a lightweight command-parsing language model using Knowledge Distillation. The main objective was to use a large teacher model train a smaller student model that can understand user commands and converts them into structured JSON response.

Task 1

I used ollama with the phi 3-5 model as the teacher model. I generated 3000 commands examples by dividing the dataset into three shards. sharding helped in distributing the workload without overlapping records.

I used seaweedfs for object storage and kubernets for running the labelling jobs.

During data cleaning 2701 records were kept out of 3000 command examples. Some records are removed because of invalid JSON.

I also uploaded both v1 & v2 model archive to the GitHub Release

Task 2

The project includes a performance evaluation setup for measuring latency, throughput, CPU usage, memory usage and estimated cost.

However, I could not complete the full benchmark performance measurement and cost analysis during this implementation. Therefore, I have not reported the measured latency, throughput or cost values.

This is an important area for the future improvement.

Challenges and learning

During this project I faced several challenges related to Python dependencies, Docker image size, Kubernetes development, model loading and network connectivity.

I learned how to debug errors, manage containers, deploy machine learning models and test application endpoints

I also used AI tools for understanding concepts, project scaffolding and debugging. I verified the generated suggestions by checking the codes and running actual tests.

future Improvement.

In the future, I would like to improve the model accuracy, reduce model size through quantization, perform completed load testing and deploy the application on the multi-node kubernetes cluster.

Conclusion

This project helped me understand the complete machine learning deployment process, starting from teacher data generation and knowledge distillation to containerization, kubernetes deployment, network security and automated testing.

It improved my practical knowledge of cloud computing, machine ~~learning~~ learning and Dev Ops.